Project Proposal

Human-Computer Interaction
Class session number: 1
Team number: 4
Team members: Yuyoung KIM, Youngbeen Kim, Jiyun Lee, Juhwan Lee, Euijin Choi

You can just use this table form for the proposal. Or you can use another format (e.g., HCI proposal previous years) containing the contents in this template. Delete blue texts when you submit the proposal.

Title	Transfer Helper (XR color route)
Abstract	Transfer Helper is an XR (Extended Reality) indoor navigation application designed to help users find their way through the complex Seoul subway system. The application renders color-coded virtual path lines overlaid on the physical floor of subway stations, guiding users to the correct platform based on their selected destination. The system directly addresses the critical pain points of missed transfers, wrong platform boarding, and the confusion experienced by both domestic commuters and foreign travelers unfamiliar with Korean subway infrastructure.
Target Users	• First-time users of the Seoul subway who are unfamiliar with complex interchange stations • Commuters who make at least one transfer on their journey and are at risk of missing a specific train • International travelers who cannot read Korean-language signage inside stations • Passengers navigating branching lines (e.g., Seoul Metro Line 1) where a single station may serve multiple platforms serving different terminal destinations
Problem	The Seoul subway network is one of the most extensive urban rail systems in the world, yet its internal complexity frequently causes confusion even for regular riders. Several structural and operational issues compound this problem: **1. Physical Complexity of Station Infrastructure** Major interchange stations such as Yongsan, Sindorim, and Guro feature multiple platforms that, despite belonging to the same line number, serve entirely different terminal destinations. The walls, pillars, and multi-level staircases of these underground stations obstruct sightlines and make directional signage easy to miss. A commuter who boards the wrong platform wastes at minimum 15 minutes and may miss critical connections.

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2. Line Branching and Variable Terminal Destinations
Seoul Metro Line 1 is a prime example of the branching problem. Multiple trains running on the same line diverge toward completely different terminal stations (e.g., Incheon, Suwon, Seodongtan, Gwangmyeong). The last train to a specific destination such as Hanam Geomdan-san Station may depart significantly earlier than the last train to a nearby terminus on the same line. A passenger who casually waits, believing another train will arrive, may find that the only remaining trains terminate several stops short of their destination, forcing a transfer to a night bus service with which they are equally unfamiliar.
3. Inadequacy of Existing Navigation Tools
Current solutions such as smartphone map apps (Kakao Map, Naver Map) provide route information at the macro level.They tell you which lines to take but offer no indoor, real-time spatial guidance once the passenger is inside the station. The passenger must still interpret static 2D station maps and physical overhead signs, which are often cluttered, inconsistently lit, or placed at junctions only after the decision point has already passed.
4. Barriers for Foreign Travelers
While major stations provide English signage, the density of information and the speed required to navigate a busy transfer station creates a high cognitive load for non-Korean speakers. Language-independent visual cues color, arrows, spatial lines are a more universally accessible navigation grammar than text-based instructions.
Academic research corroborates the significance of this problem. A study published in the Journal of the Korean Society of Transportation (대한교통학회지) evaluated the transfer efficiency of Seoul metropolitan rail stations and identified structural wayfinding deficiencies as a key contributor to suboptimal passenger flow. A report on Seoul students' psychological changes during COVID-19 also documented the rising importance of reducing unnecessary physical and cognitive burden in public transit environments.
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<td>Solution</td>
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The final goal of Transfer Helper is to construct an XR-based indoor navigation system that enables users to reach the correct platform in the most intuitive way possible, without getting lost inside complex subway stations.
Technical Approach

Device & Platform: The system runs on a Meta Quest 3 headset, which provides high-quality passthrough capability. The user sees the real station interior overlaid with virtual guidance content.
Spatial Anchoring via QR Markers: QR codes or image markers are placed at key positions in the station (e.g., ticket gates, major pillars). The application scans these markers to synchronize the virtual coordinate system with the real physical space, enabling accurate overlay rendering.
Path Rendering with Unity & OpenXR: Using the Unity XR Interaction Toolkit and Meta XR SDK, the application detects the real floor geometry and renders a spline-based color-coded line in the line's official color (e.g., blue for Line 1, green for Line 2) directly on the floor surface, leading the user to the correct platform.

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	● Intelligent Routing Algorithm: Based on the user's selected destination, the algorithm calculates the most efficient path prioritizing routes with fewer stairs and faster transfer times and visualizes this path as the floor guide. **Value Delivered to Users** ● Eliminates wrong-platform boarding at branching interchange stations ● Reduces physical exertion by optimizing walking route within the station ● Relieves schedule anxiety caused by missed transfers or directional confusion ● Provides language-independent navigation via color and arrow visual cues, making the system equally usable for foreign travelers regardless of Korean language proficiency
Main Functions	1. Dynamic Color-Path Mapping: The user selects a final destination, and the system renders a virtual floor line in the official color of the relevant subway line, providing the clearest possible visual route from the current position to the target platform. 2. Platform-Specific Guidance at Branching Points: At stations where the same line number serves multiple platforms for different terminal destinations (e.g., Yongsan Station), the system analyzes the user's terminal station data and navigates them to the exact correct boarding zone. 3. Haptic and Visual Off-Path Feedback: If the user deviates from the designated route or moves in the wrong direction, the system immediately provides a red warning UI overlay on the display and a vibration alert through the controller, prompting course correction. 4. Context-Aware Information Overlay: Upon reaching a transfer point or target platform, the system displays real-time contextual information such as estimated arrival time of the next train and current train crowding level as an XR interface element floating in the physical space.
Interaction Concepts	● Destination selection: Pinch gesture in mid-air to select from a floating radial menu ● Tower rotation equivalent: Head gaze + dwell-time selection for accessibility fallback ● On-path confirmation: Floor color line renders and pulsates to confirm active guidance ● Off-path alert: Red overlay flash + haptic buzz on controller ● Arrival confirmation: Information panel auto-appears at platform using spatial anchor trigger

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<td>Plan of Interactions</td>
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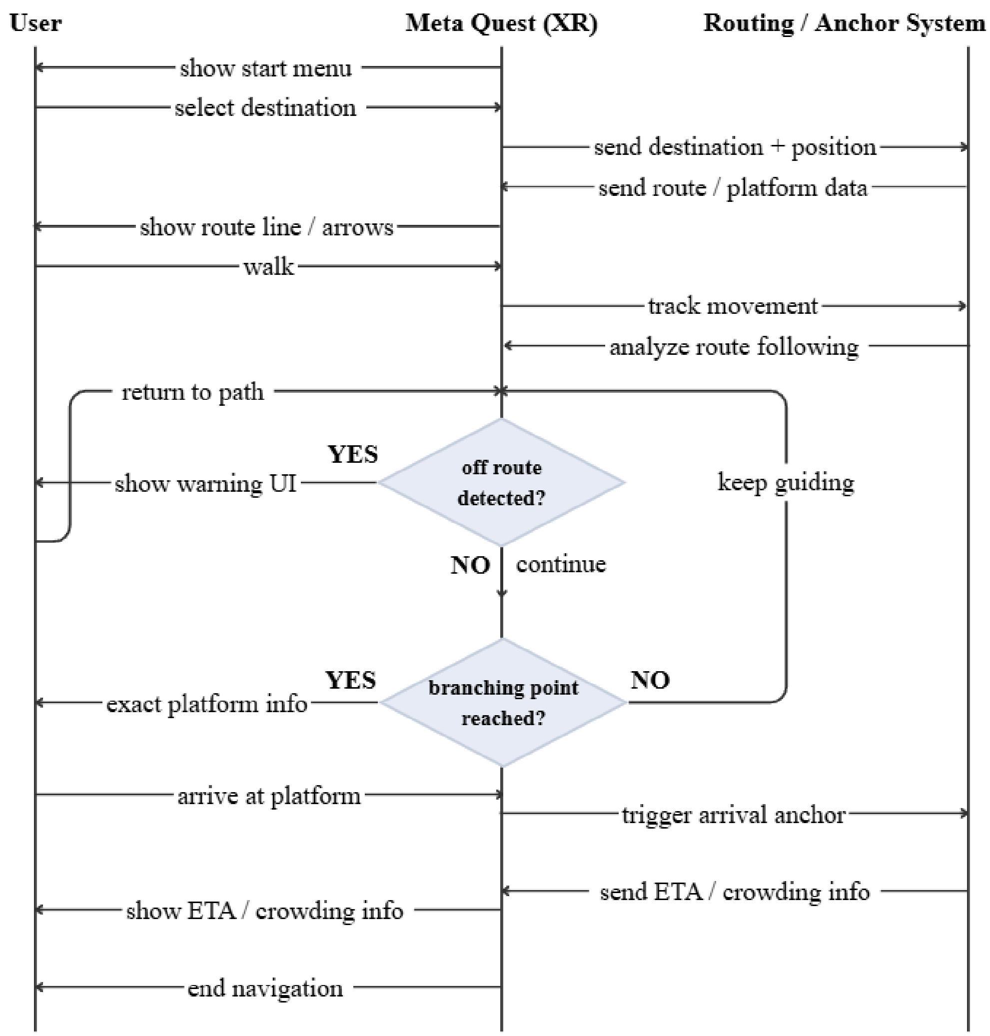

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<td>Implementation</td>
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<table>
<tr><td>List of requirements</td><td>Details</td></tr>
<tr><td>Programming language</td><td>C#, C++</td></tr>
<tr><td>Source codes, examples</td><td>Meta Quest Hello World example
Meta Passthrough tutorial examples
Unity OpenXR / Meta Mixed Reality scene setup examples
Unity XR Origin basic examples</td></tr>
<tr><td>APIs, libraries</td><td>Unity OpenXR Plugin
Unity OpenXR: Meta
Meta XR Core SDK
Meta XR Interaction SDK (if needed)</td></tr>
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		XR Interaction Toolkit (if needed)
	Device	Meta Quest 3 headset PC or laptop for development
	IDE	Unity Editor, Visual Studio (C# code editor)
	Additional software	Android Build Support OpenJDK Android SDK and NDK Tools XR Plug-in Management Meta Quest mobile app for enabling Developer Mode
	Media sources (images, videos, music)	Arrow icons for navigation Destination icons QR codes or image markers for starting point recognition Indoor map or floor plan images Simple sound effects or voice guidance if needed
	Data	Route data from the classroom to the restroom or bus stop Waypoint coordinate data for the path Location data for the starting point and destination Marker reference location data Optional labels for hallways, exits, or floors

Schedule

Academic Week	**Main Goal**	**Detailed Tasks**
Week 4	**Finalizing the project plan**	Confirm the project topic, define the MVP scope, decide the starting point and destination, organize team roles, and finalize the main features and excluded features
Week 5	**Setting up the development environment and deciding versions**	Choose the Unity version, set up the Meta Quest 3 development environment, install OpenXR and Meta XR SDK, test Quest 3 connection, and organize GitHub or collaboration tools

Week 6	**Implementing the basic XR environment**	Set up passthrough, configure XR Origin, create the basic scene, display simple virtual objects in the real environment, and build a simple start UI
Week 7	**Implementing the navigation function**	Define the starting position method, create waypoint-based paths, display navigation lines or arrows, implement destination selection, and connect different routes to different destinations
Week 8	**Stabilizing the prototype and preparing for testing**	Improve visibility of arrows and path lines, adjust object position and size, add start/restart buttons, fix bugs, prepare a demo flow, and create user testing questions
Week 9	**User Testing I and feedback collection**	Conduct the first user test, observe usability issues, collect feedback on visibility, clarity, and ease of use, and organize improvement priorities
Week 10	**Code revision based on feedback**	Modify the UI, improve the navigation visuals, refine object placement, simplify the interaction flow, and stabilize the system performance
Week 11	**User Testing II, feedback collection, and final revisions**	Conduct the second user test, validate the system, collect final feedback, and apply the final revisions to improve usability and system quality
Week 12	**Video recording**	Finalize the demo scenario, record the system demonstration video, capture screenshots, and prepare presentation materials

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<tr><td>Final Outputs</td><td>A Meta Quest 3-based XR indoor navigation MVP

Destination selection function for places such as the restroom or bus stop

Basic XR interface including a start menu and simple interaction flow

A demo video showing the main features and usage of the system

A final presentation explaining the project background, development process, and results

A final report documenting the design, implementation, and evaluation of the system

User testing results including feedback and revisions

Optional: a dedicated support app to help users with destination selection, instructions, and simple guidance before using the XR system</td></tr>
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Submission

- Page limit: minimum of 7 pages (including figures)
- Format: Times New Roman, 11 pt., 1.15 line space
- File format: PDF
- Language: English
- Due: announced at LMS
- Submission to LMS

Evaluation

- Detail of planning
- Feasibility

Sample Proposals

https://drive.google.com/drive/u/0/folders/1PIkLDID3gr3pYDB0nDtpLI7cIpL-kn7S